1 **Install the CPU onto the motherboard**

- Align the CPU with the socket markers and lock it into place.
- The motherboard is easiest to work on while it is outside the case and unobstructed.

2 **Install the CPU cooler**

- Apply thermal paste if needed and mount the cooler.
- Connect the cooler fan to the CPU_FAN header.
- Access to the CPU area is much easier before the motherboard is inside the case.

3 **Install RAM modules**

- Insert the memory sticks into the correct slots until the latches click.
- Large coolers and tight cases can make RAM installation harder later.

4 **Install M.2 SSDs (if present)**

- Insert the drive into the M.2 slot and secure it.
- Some M.2 slots become difficult to reach once the motherboard is mounted.

5 **Install the power supply (PSU)**

- Secure the PSU in its mounting location.
- Route major cables toward their destinations.
- It provides the power connections needed for the rest of the build.

6 **Connect motherboard power cables**

- Attach the 24-pin motherboard connector.
- Attach the CPU power connector (typically 4+4 or 8-pin).
- These cables can be difficult to route once other hardware is installed.

7 **Install storage drives (2.5-inch or 3.5-inch)**

- Mount drives in their bays.
- Connect SATA power and data cables if required.
- The motherboard and PSU are already in place, making cable routing straightforward.

8 **Install the graphics card (GPU)**

- Insert it into the primary PCIe slot and secure it.
- Connect PCIe power cables if required.
- The GPU depends on the motherboard and often occupies significant space.

9 **Connect front-panel cables**

- Power switch, reset switch, power LED, HDD LED.
- Front USB and audio connectors.
- Small connectors are easier to manage once component positions are fixed.

10 **Final inspection**

- Verify all power connectors are fully seated.
- Check RAM, GPU, storage, and fan connections.
- Ensure no loose screws remain inside the case.
- Prevents common startup and hardware issues.

11 **Power-on test**

- Connect monitor, keyboard, and power.
- Start the system and confirm it reaches BIOS/UEFI.
- Check that fans spin and drives/RAM are detected.
- Confirms the entire assembly was completed correctly.